

Systems Analysis and Design

Chapter 1

Introduction to System Analysis & Design(SAD)

Learning Objectives

- ✓ Define information systems analysis and design
- ✓ Discuss the modern approach to SAD that combines process and data views
- ✓ Describe the SDLC and alternatives
- ✓ Describe role of computer aided software engineering (CASE) tools

What is Information Systems Analysis and Design?

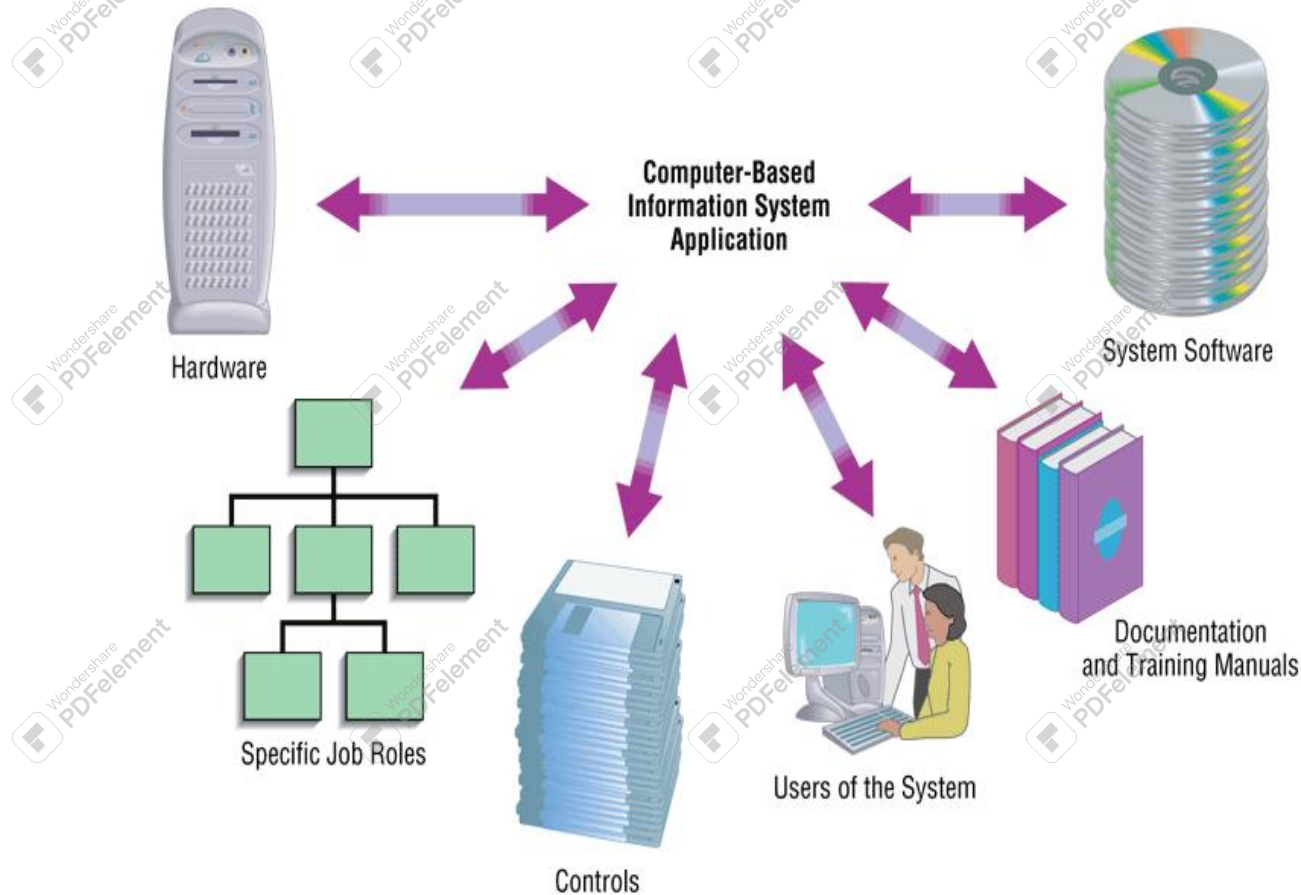
- A method used to **create and maintain systems** that perform basic business functions.
- Main goal - to **improve employee efficiency by applying software solutions** to key business tasks
- A structured approach to ensure success
- Systems Analysts perform SAD based upon:
 - Understanding of organization's objectives, structure and processes
 - Knowledge of how to exploit information technology for advantage

Systems Analysis and Design: Core Concepts

- Major goal: to improve organizational systems by developing or acquiring application software and training employees in its use
- System: Turns data into information and includes:
 - Hardware and system software
 - Documentation and training materials
 - Job roles associated with the system
 - Controls to prevent theft or fraud
 - The people who use the software to perform their jobs

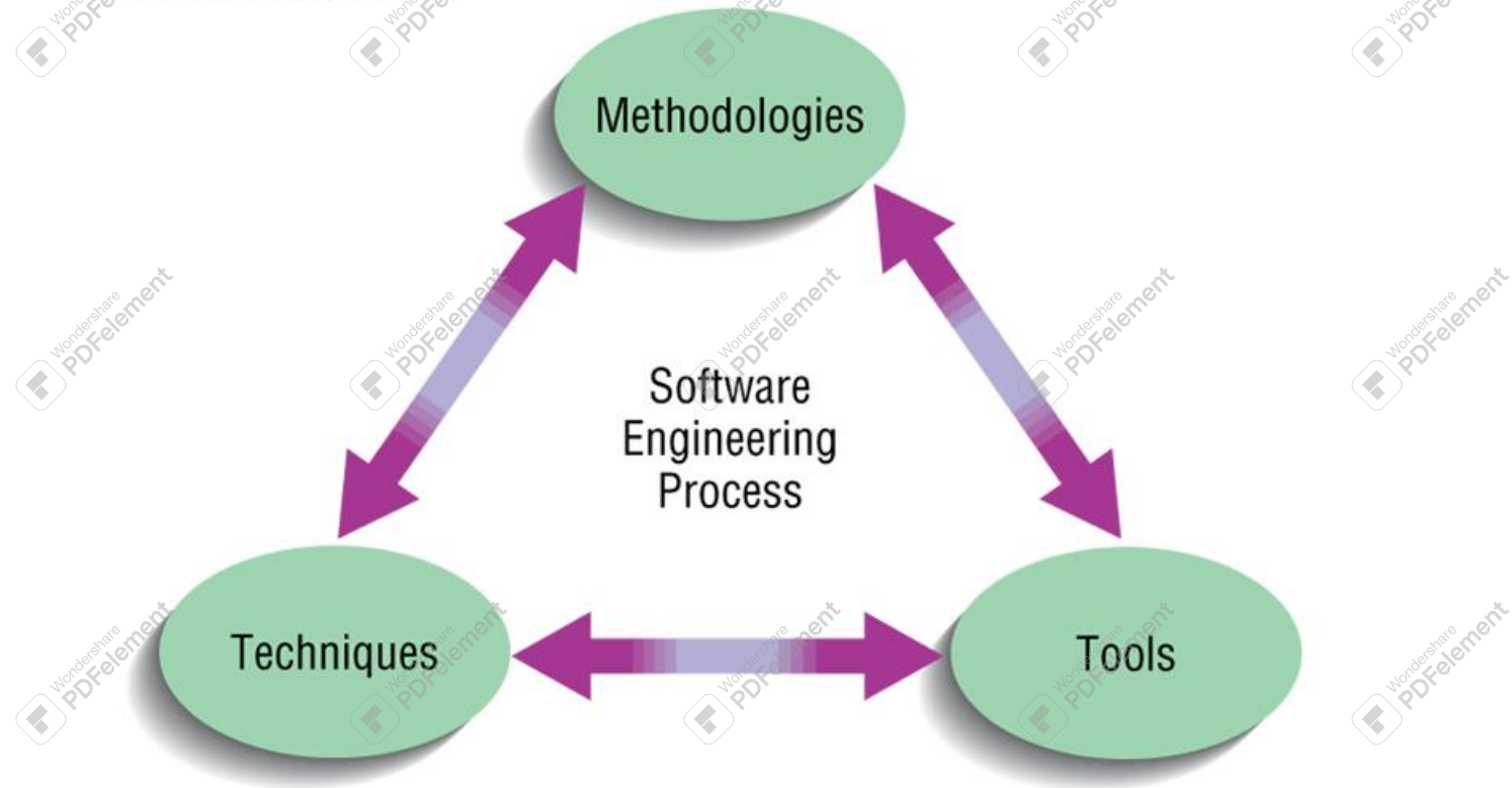
Systems Analysis and Design: Core Concepts

Figure 1.2 Components of a Computer-Based Information System Application



Software Engineering Process

Figure 1.3 The software engineering process uses methodologies, techniques, and tools.



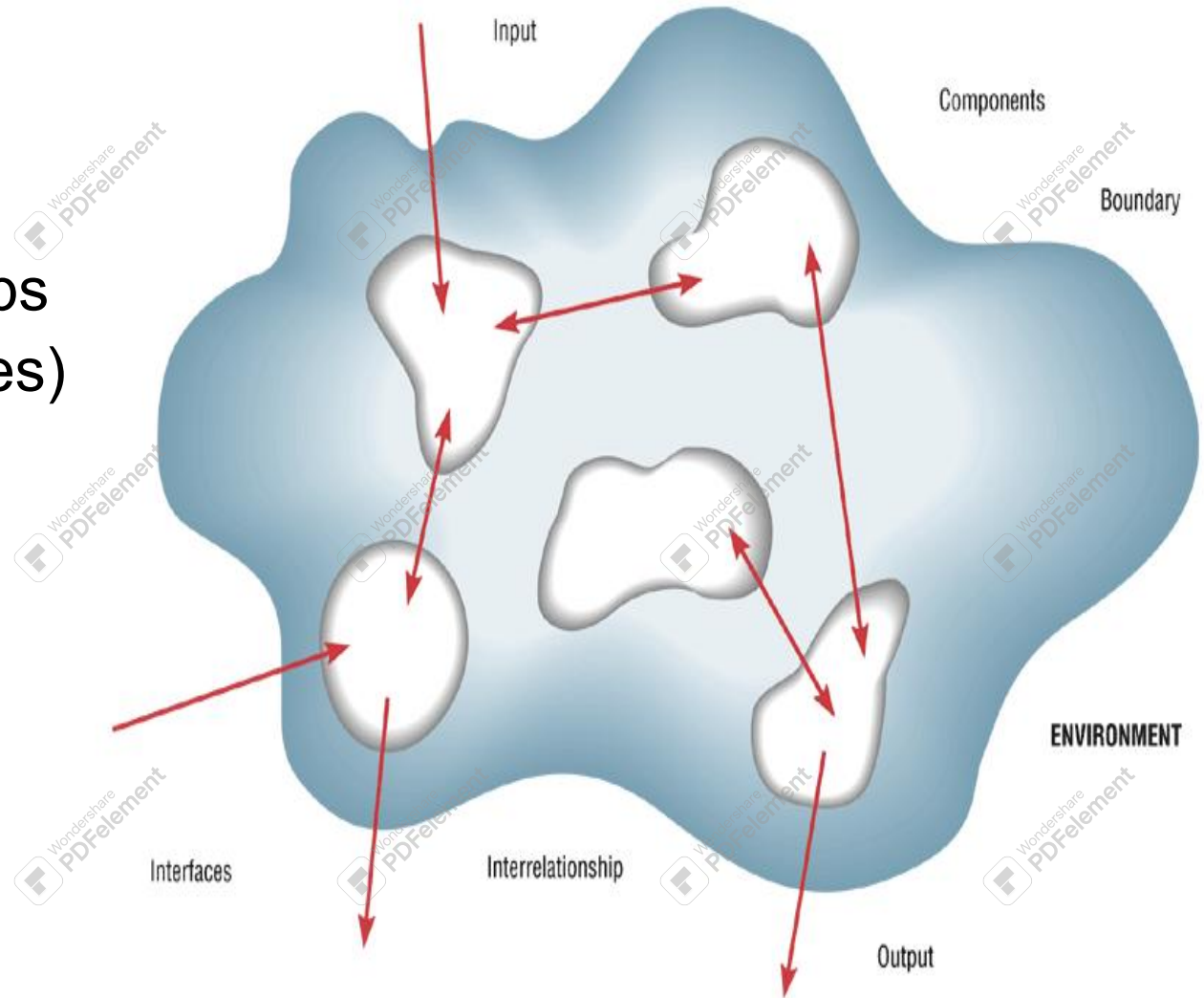
System

- A system is an interrelated set of business procedures used within one business unit working together for a purpose
- A system exists within an environment
- A boundary separates a system from its environment
- A system has nine characteristics

Characteristics of a System

Figure 1.4 Seven Characteristics of a System

- > Purpose
- > Environment
- > Boundary
- > Components
- > Interrelationships
- > Constraints(rules)
- > Interfaces
- > Input
- > Output



Important System Concepts

- **Decomposition**
 - The process of breaking down a system into smaller components
 - Allows the systems analyst to:
 - Break a system into small, manageable and understandable subsystems
 - Focus on one area at a time, without interference from other areas
 - Concentrate on component pertinent to one group of users without confusing users with unnecessary details
 - Build different components at independent times and have the help of different analysts

Important System Concepts

- Modularity
 - Process of dividing a system into modules of a relatively uniform size
 - Modules simplify system design
- Coupling
 - Subsystems that are dependent upon each other are coupled
 - Desired: loose coupling
- Cohesion
 - Extent to which a subsystem performs a single function
 - Desired: high cohesion

Cohesion and Coupling

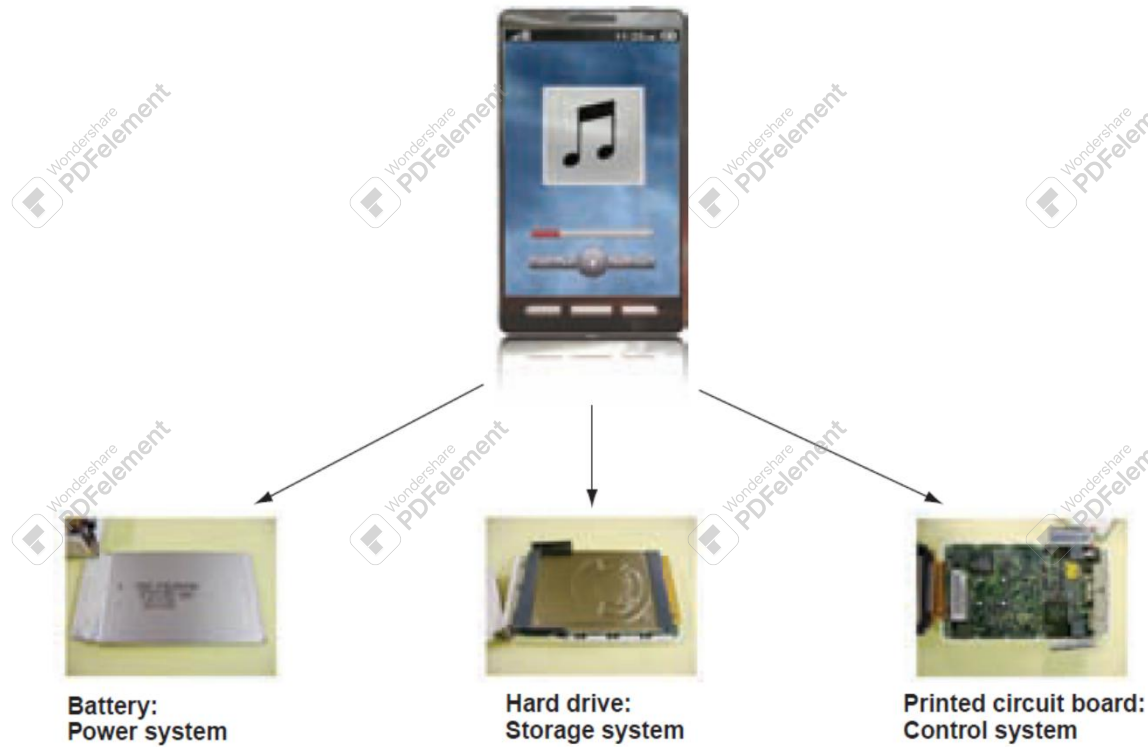


FIGURE 1-6

An MP3 player is a system with power supply, storage and control subsystems.

Sources: Shutterstock; ©Harald van Arkel/Chipmunk International;

A Modern Approach to Systems Analysis and Design

- Systems Integration
 - Allows hardware and software from different vendors to work together
 - Enables procedural language systems to work with visual programming systems
 - Visual programming environment uses client/server model

SA role in Systems Development

- Study problems and needs of an organization
- Determine the best approach to improve organization through use of:
 - People
 - Methods
 - Information technology
- Help system users and managers define their requirements for new or enhanced information systems

Systems Development Methodology ... e.g. SDLC

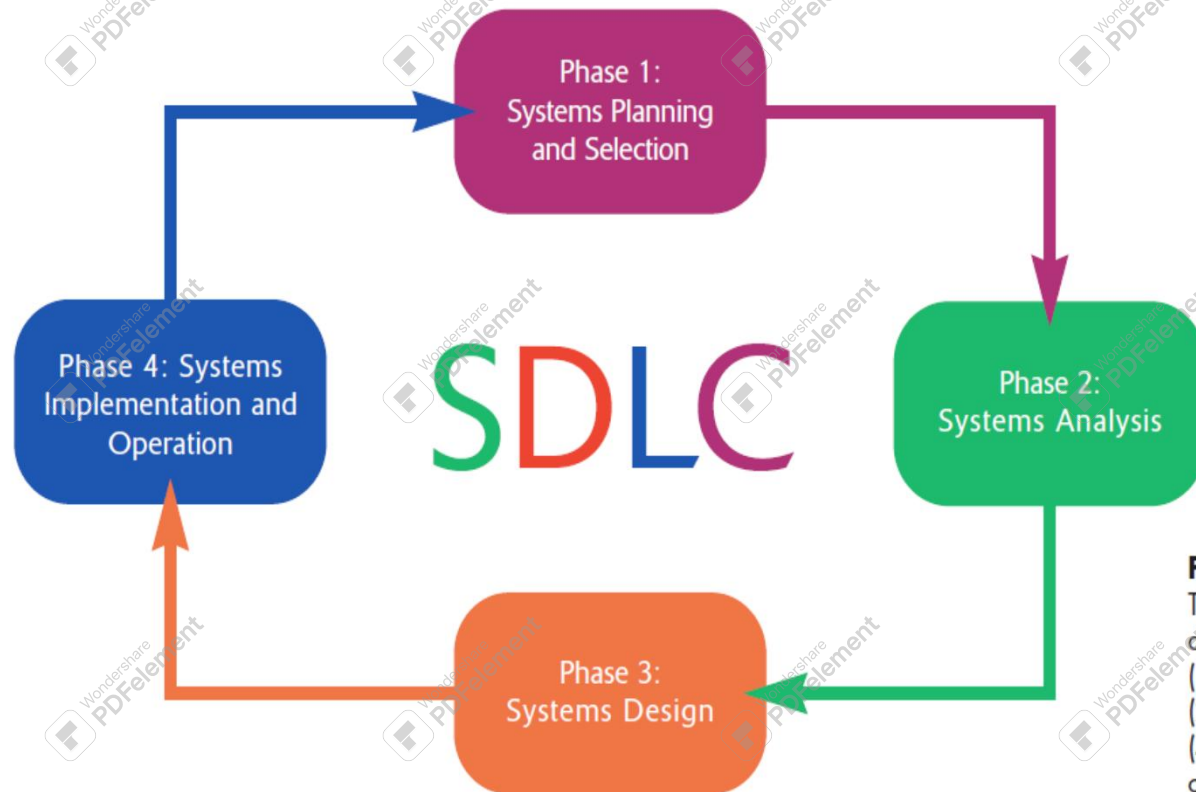


FIGURE 1-1

The four steps of the systems development life cycle (SDLC): (1) planning and selection, (2) analysis, (3) design, and (4) implementation and operation.

Phases of the Systems Development Life Cycle

1. Systems Planning and Selection

- Two Main Activities
 - Identification of need
 - Investigation and determination of scope

2. Systems Analysis

- Study of current procedures and information systems
 - Determine requirements
 - Generate alternative designs
 - Compare alternatives
 - Recommend best alternative

Phases of the Systems Development Life Cycle

3. System Design

- Logical Design - Business aspects of the system
- Physical Design - Technical specifications

4. System Implementation, Operation and Maintenance

- Hardware and software installation
- User Training
- Documentation
- Operations
- Maintenance (fix errors, make changes)

Alternative Approaches to Development

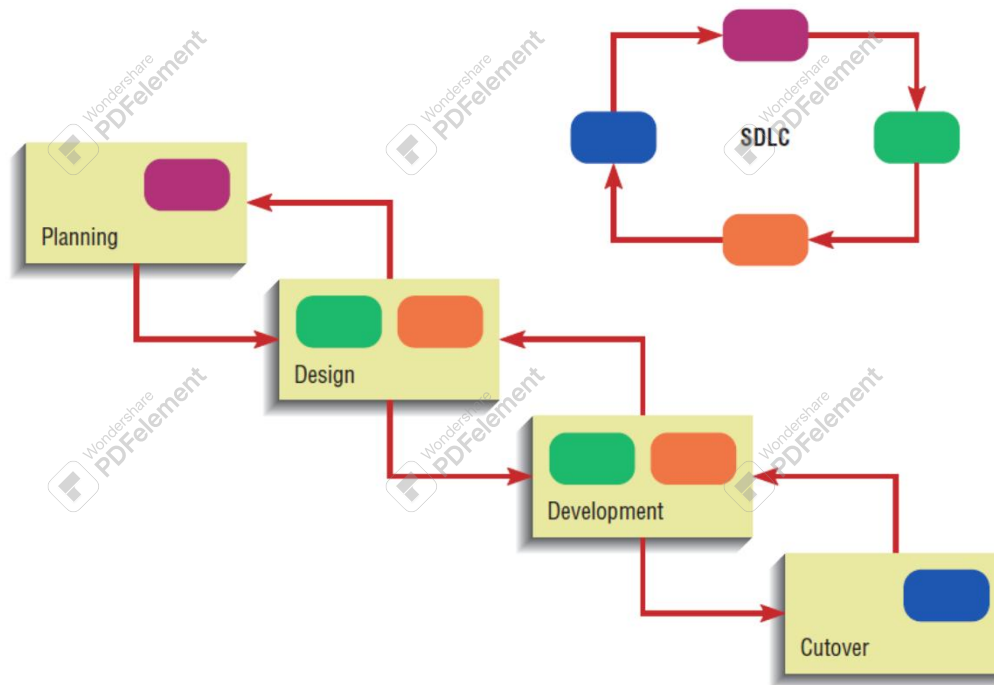
- Prototyping
 - Building a scaled-down working version of the system
 - Advantages:
 - Users are involved in design
 - Captures requirements in concrete form
 - Disadvantages:
 - Users may misunderstand SD complexity
 - Code often thrown away

Alternative Approaches to Development

- Joint Application Design (JAD)
 - Users, Managers and Analysts work together for several days
 - System requirements are reviewed
 - Structured meetings
- Rapid Application Development (RAD)
 - Utilizes prototyping to delay producing system design **until user requirements are clear**

FIGURE 1-13

RAD systems development life cycle compared to standard SDLC.



Approaches to Development

- Agile Methodologies
 - Extreme Programming, Adaptive S/W Development, Scrum
 - Focuses on
 - Adaptive methodologies
 - People instead of roles
 - Self-adaptive development process

Computer -Aided Software Engineering (CASE) Tools

- Automated software tools used by systems analysts to develop information systems
 - Project management tools (Openproj, MS Project)
 - Diagramming tools (Dia, Visio)
 - Computer display and report generators (VB)
 - Analysis tools (Visual Analyst)
 - Documentation generators
 - Code generators
 - Integration of tools via a repository